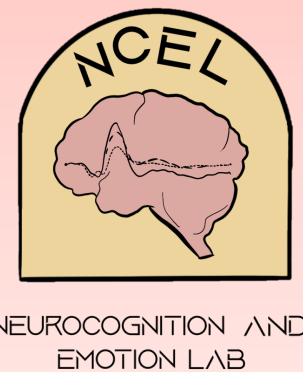


# Seeing Negativity in Neutral Faces: Trauma-Related Symptoms and Interpretation Bias

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## Background

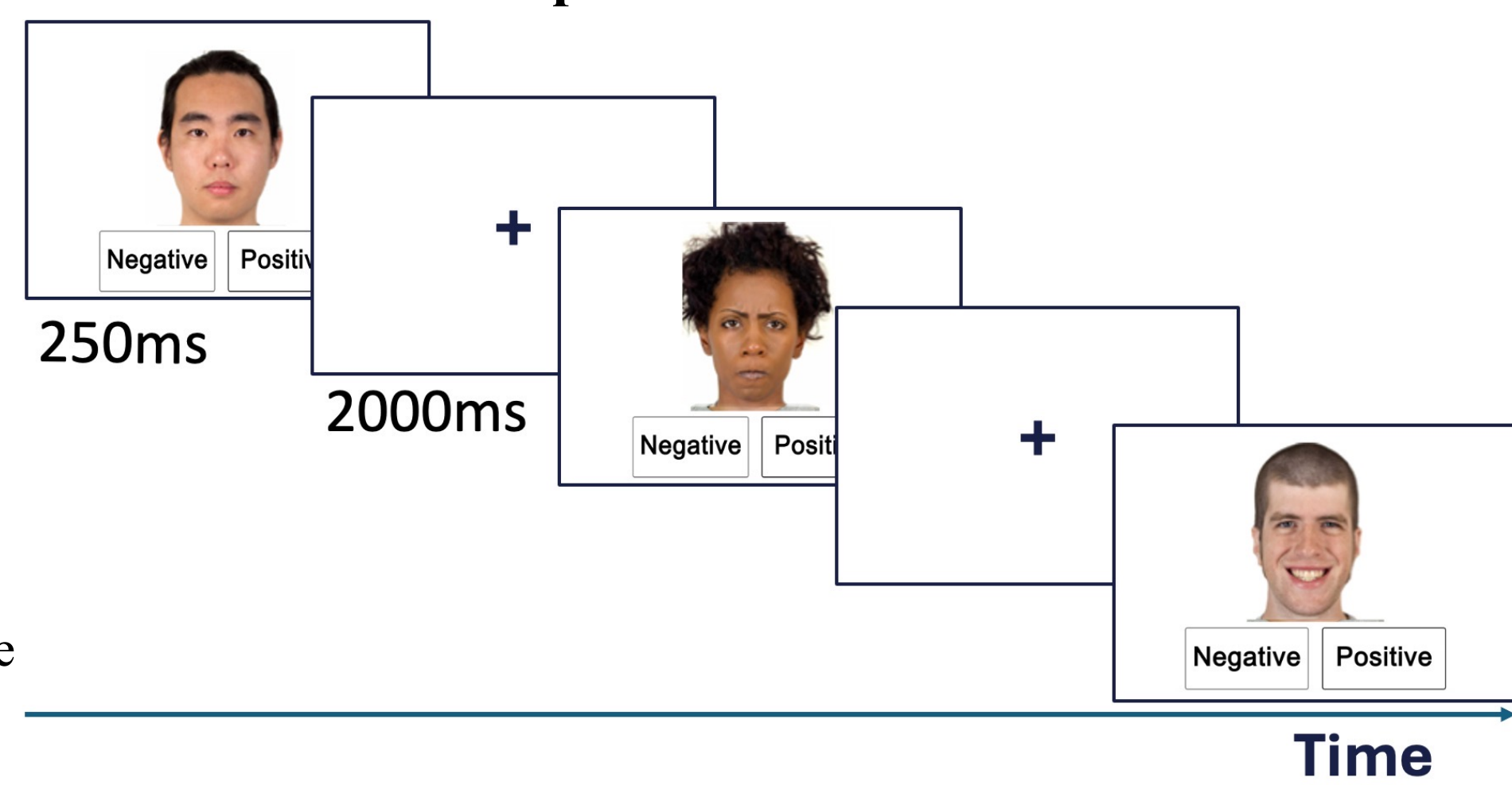
- Trauma-related psychopathology may have lasting effects on socioemotional processing, including the interpretation of emotional cues (Ehlers & Clark, 2000; Poljac et al., 2011).
- Neutral facial expressions do not clearly signal a specific emotion, allowing examination of responses to subtle and ambiguous facial information.
- The effects of trauma may extend beyond directly exposed individuals, potentially influencing their offspring's mental health and socioemotional functioning (Dekel & Goldblatt, 2008; Leen-Feldner et al., 2013).
- In WTC-responder families, greater parental 9/11 exposure was associated with PTSD, anxiety, and panic symptoms among adult offspring; parental PTSD and depression were also associated with corresponding symptoms in offspring (Cykowicz et al., 2026).

## Objectives

To examine emotional processing in WTC responder parents and their adult offspring in relation to current mental health symptoms and parental 9/11 exposures. We hypothesized that greater current symptom severity and greater parental exposure would be associated with a greater tendency to interpret neutral faces negatively.

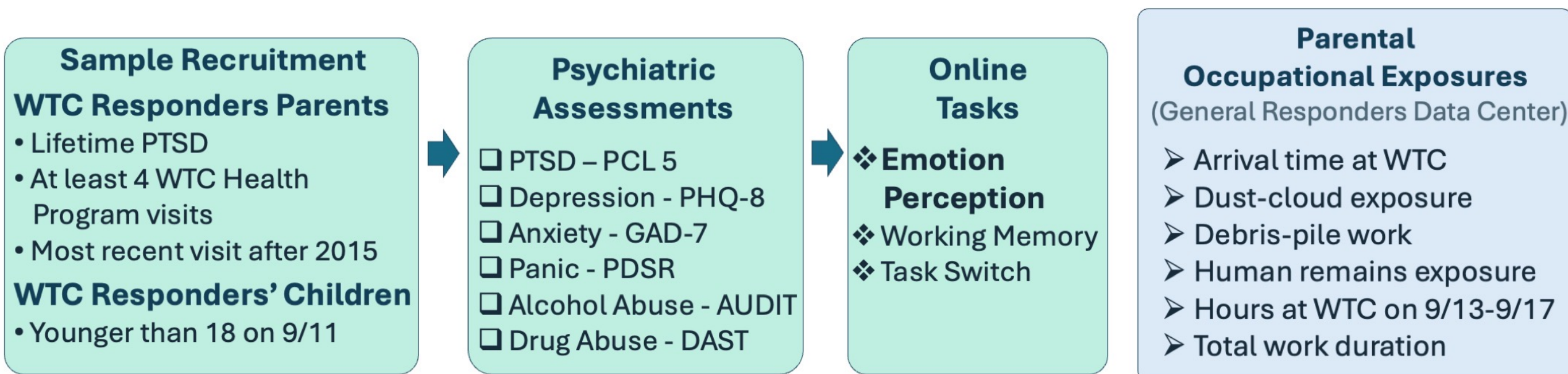
## Study Design

### Facial Emotion Perception Task



Valid Responses:

- Positive to Neutral Face
- Negative to Neutral Face
- Positive to Positive Face
- Negative to Negative Face



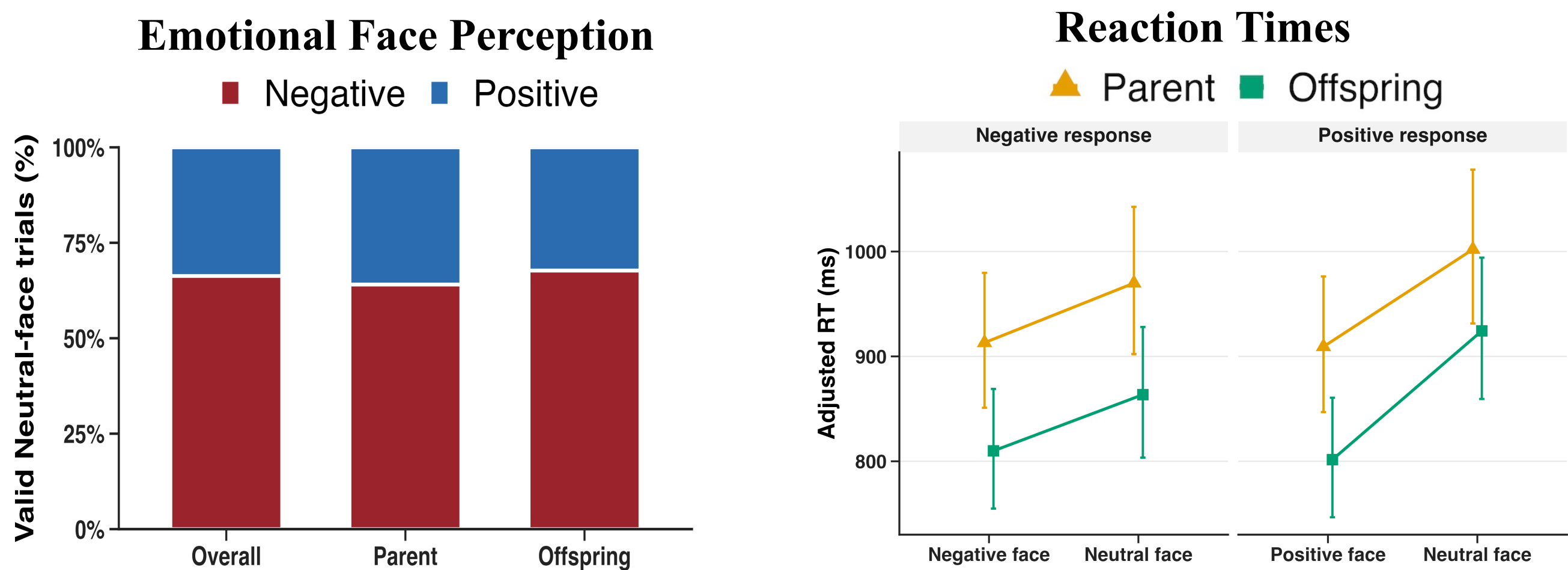
## Methods

- Mental-health symptoms were examined continuously, with AUDIT and DAST included in sensitivity analyses.
- Neutral-Positive and Neutral-Negative responses were analyzed using binomial mixed-effects models with random intercepts for family, participant, and facial stimulus.
- Linear mixed-effects models were used to compare log-transformed reaction times (RTs) across response types.
- Models were adjusted for generation, age, sex, RTs, and device.

## Results

### Neutral-Face Emotional Perception and Reaction Time to Emotional Faces

- Both parents and children showed a bias toward judging neutral faces as negative.
- RT to neutral faces were slower than RTs to either positive or negative faces. ( $p < .001$ ), whereas RTs to positive and negative faces did not differ ( $p = .419$ ).
- For neutral faces, negative judgments were 4.9% faster than positive judgments ( $p < .001$ ), with a significant difference among Offspring ( $p = .008$ ).



### Association of Face Emotion Interpretation and Parental WTC Exposure

- Greater parental hours at the WTC-site during 9/13-9/17 were associated with more negative neutral faces judgments among offspring only ( $OR = 1.43$ ,  $p = .003$ ), with significant generation difference ( $p = .005$ ).
- Human-remains exposure was associated with more negative judgements among offspring ( $OR = 1.71$ ,  $p = .043$ )

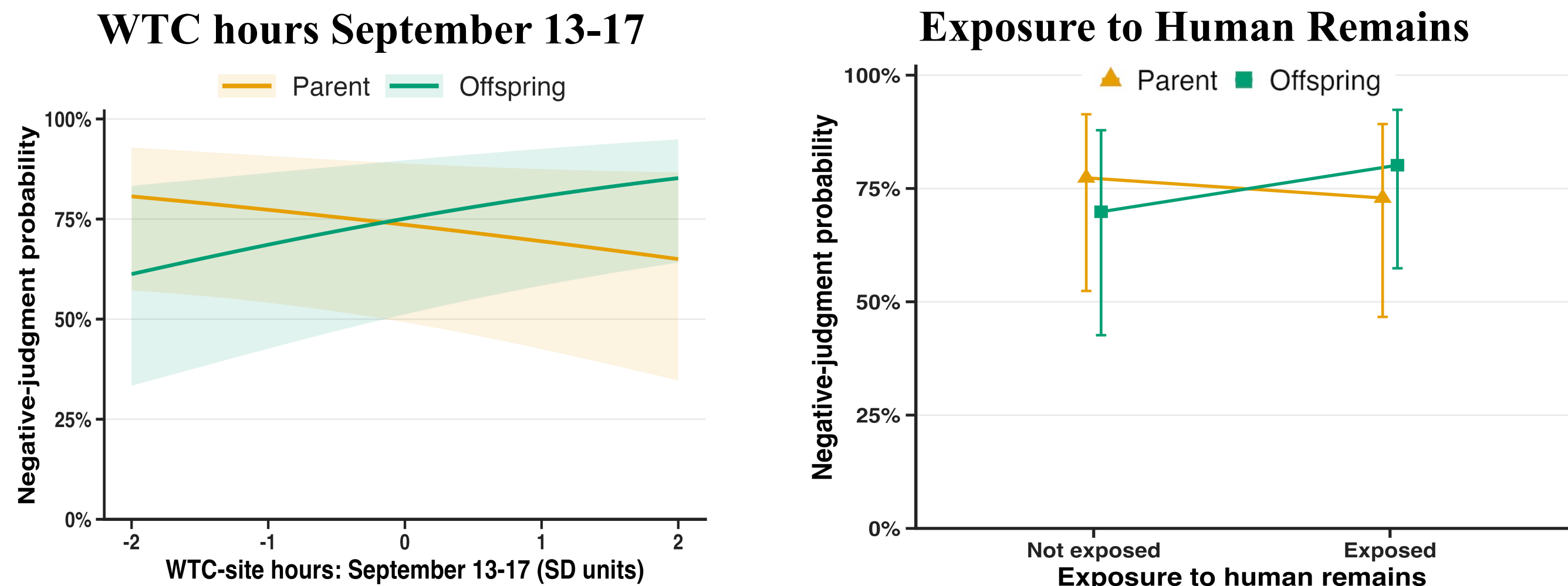
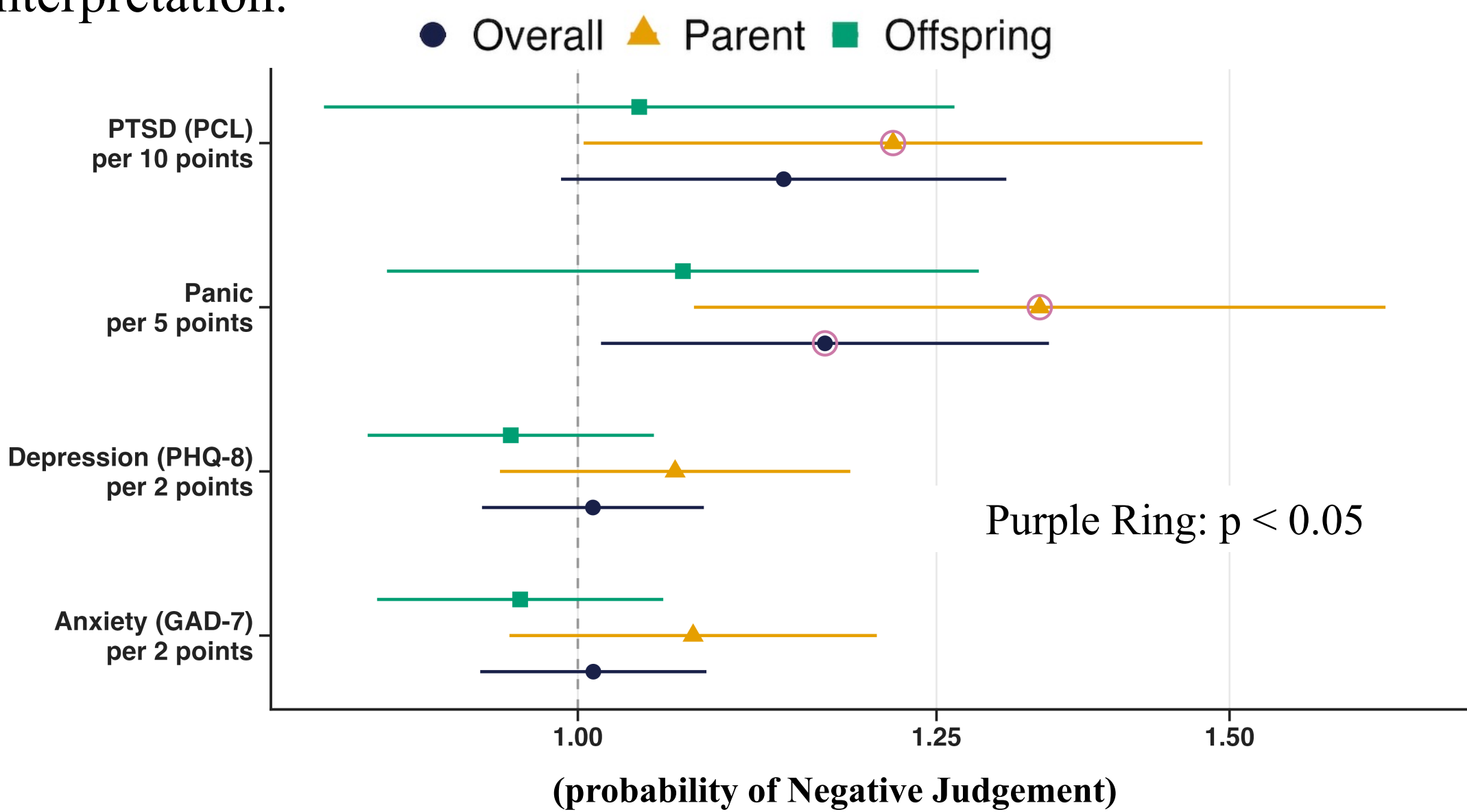


Table 1. Participants' Demographics

| Characteristic            | Overall (N=413) | Parent (N=165) | Offspring (N=248) |
|---------------------------|-----------------|----------------|-------------------|
| Age, M (SD), years        | 41.3 (16.3)     | 60.1 (5.7)     | 28.7 (5.1)        |
| Female, n (%)             | 173 (42.4)      | 20 (12.1)      | 153 (63.0)        |
| Spanish speaking, n (%)   | 101 (24.9)      | 29 (17.8)      | 72 (29.6)         |
| Race and ethnicity, n (%) |                 |                |                   |
| Black, non-Hispanic       | 12 (3.0)        | 6 (3.8)        | 6 (2.5)           |
| Black, Hispanic           | 12 (3.0)        | 7 (4.5)        | 5 (2.1)           |
| White, non-Hispanic       | 267 (67.3)      | 110 (70.5)     | 157 (65.1)        |
| White, Hispanic           | 56 (14.1)       | 18 (11.5)      | 38 (15.8)         |
| Minoritized groups        | 50 (12.6)       | 15 (9.6)       | 35 (14.5)         |

### Facial Emotion Interpretation and Psychological Symptoms

- Panic symptoms were associated with more negative interpretations overall ( $OR = 1.17$ ,  $p = .031$ ), with a stronger association among parents ( $OR = 1.33$ ,  $p = .009$ ), but not offspring ( $p = .487$ ).
- PTSD symptoms showed a similar pattern, with significant relationship among Parents ( $OR = 1.22$ ,  $p = .046$ ), and approaching significance for overall ( $p = .070$ ).
- Depression and generalized anxiety were not associated with negative interpretation.



## Conclusions

- Neutral faces were more often interpreted as negative than positive, with similar interpretation probabilities across generations.
- Greater panic and PTSD symptoms were associated with higher frequency of negative interpretations, particularly among parents, linking current symptoms to processing ambiguous facial expressions.
- Greater parental WTC exposure, including time at the site and exposure to human remains, was associated with more negative interpretations among offspring, suggesting an intergenerational influence on emotional processing.
- Positive and negative faces elicited faster responses than neutral faces, demonstrating the ambiguity in decision. For neutral faces, negative judgments were faster than Positive judgments, particularly among offspring, demonstrating greater bias towards negative interpretations.
- Overall, parental trauma exposure may shape offspring emotional information processing, suggesting a potential cognitive pathway for intergenerational effects of trauma.

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